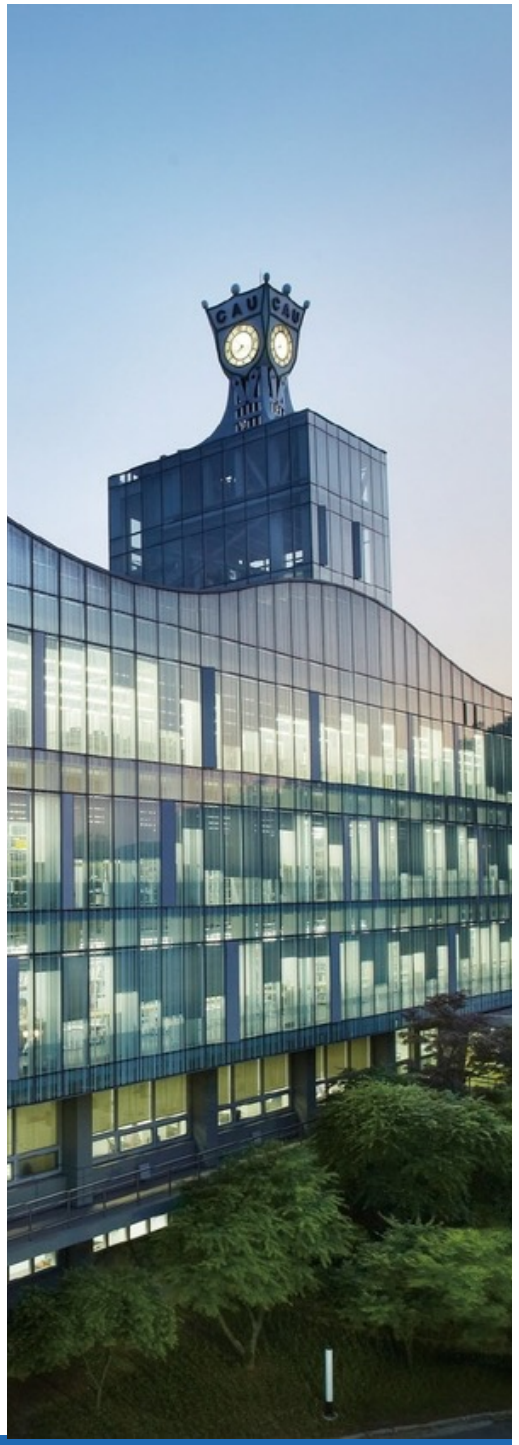
Welcome to Principles of Programming Languages

Lecture 1
Course Overview

Jinsung Kim, Ph.D.
High-Performance Computing (HPC) Lab
Computer Science and Engineering
Chung-Ang University



Course Overview

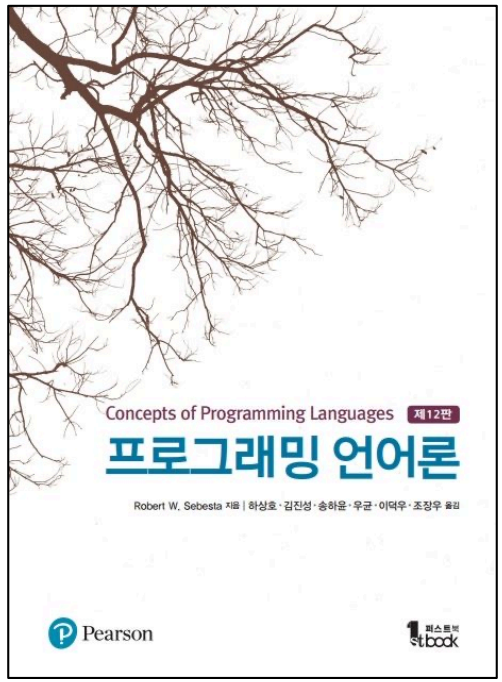
- **Instructor: Jinsung Kim**
  - Research areas: High-Performance Computing & Deep Learning
  - Office: Room 505, Bldg. 208
  - Office Hours: Tue & Thu 5PM-6PM
  - Email: kimjsung@cau.ac.kr

- **Course Title**
  - Principles of Programming Languages (프로그래밍 언어론)

- **Hours and Location**
  - 01: Room 723, Bldg. 310, Tue 11:00-13:00, Thu 12:00-13:00 (729)
  - 02: Room 723, Bldg. 310, Tue 13:00-15:00, Thu 13:00-14:00 (729)

Course Overview

- **Textbook**
  - Sebesta, Robert W. Concepts of programming languages. Pearson
- **Prerequisites**
  - This course assume that each student is an expert in one or more high-level programming languages
    - e.g., C/C++, Java, Python



- **Course Description**
  - The fundamental concepts of computer programming languages
  - Formal methods used to describe the syntax and semantics of programming languages, implementation techniques for various language constructs, and implementation of subprogram linkage

Course Overview

- **Course Objectivies**
  - Understand core principles
    - Syntax, semantics, types, scoping, control structures
  - Compare major paradigms
    - Imperative, functional, and logic programming
  - Gain hands-on experience
    - Implement a mini language (syntax and semantics)
  - Reason about language design
    - Trade-off in readability, writability, and etc.
  - Connect to modern platform
    - See how paradigms appear in GPU programming and frameworks like PyTorch

Course Overview

- **Syllabus**
  - Ch.1: Preliminaries
  - Ch.3: Describing Syntax and Semantics
  - Ch.4: Lexical and Syntax Analysis
  - Ch.5: Names, Bindings, and Scopes
  - Ch.6: Data Types
  - Ch.7: Expressions and Assignment Statements
  - Ch.8: Statement-Level Control Structures
  - Ch.9: Subprograms
  - Ch.10: Implementing Subprograms
  - Ch.13: Concurrency
  - Ch.15: Functional Programming Languages

Course Overview

- **Grading Policy**

Assessment Items	%
Attendance / Participation	10%
Mid-term Exam	35%
Assignments / Projects	20%
Final Exam	35%
Total	100%

- **F Grade**
  - Any student who is absent for one or more of thw two scheduled exams will receive an F grade
  - Any student with 4 or more absences or 10 or more latenesses througout the course will receive an F grade
- **NO CLASS THIS THURSDAY (Sep. 3)!**
- **(TBD) Online Courses: Nov. 5, 10, 12**

Q & A